

Econ 2 - Lecture 7 - 4/20/26

Midterm Exam next Monday, 4/27/26

Chapters 1, 2, 3 (up to unemployment)

Head TA will send out a seating chart on Canvas

Bring a calculator + pen/pencil

Equations given for: Growth Rate, CPI, weighted basket cost,
GDP Deflator, Unemployment Rate

Today: Finish full Employment (Chapter 3.3)

Wednesday: Most of lecture material NOT on Exam

Lecture Quiz #4 Released Wednesday, due Monday, 4/27

Questions on Quiz = Exam Prep

Last Class: Categories of Unemployment

Frictional Unemployment: time lag in hiring process

⇒ Voluntary leave → search for a new job

↳ New job is not instant

Seasonal Unemployment: predictable patterns in employment

⇒ Sports Coach, Landscaping, etc.

⇒ Neither frictional or seasonal unemployment
represent the larger economy

Structural Unemployment

Mismatch between skills of workers and employer needs

Movement from analog to digital phone services

→ Switchboard operators in mid-1950s → not needed in 1990s

→ Potential long-term job loss

→ New demand for programmers

Cost to some workers who do not have the appropriate skills for future work

Likely increase in productivity

Frictional + Seasonal + Structural Unemployment

Natural Unemployment

Expect Natural Rate of Unemployment $> 0\%$

Cyclical Unemployment:

Unemployment that is caused by a direct change in output/production (Y)

Example: US Car industry in 2009

High gas prices + Housing Crash → Lower Demand for

Layoffs

← US Car Companies

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Gas-Heavy Vehicles

→ Trucks + SUVs

Buying fewer Trucks → Layoffs at Ford

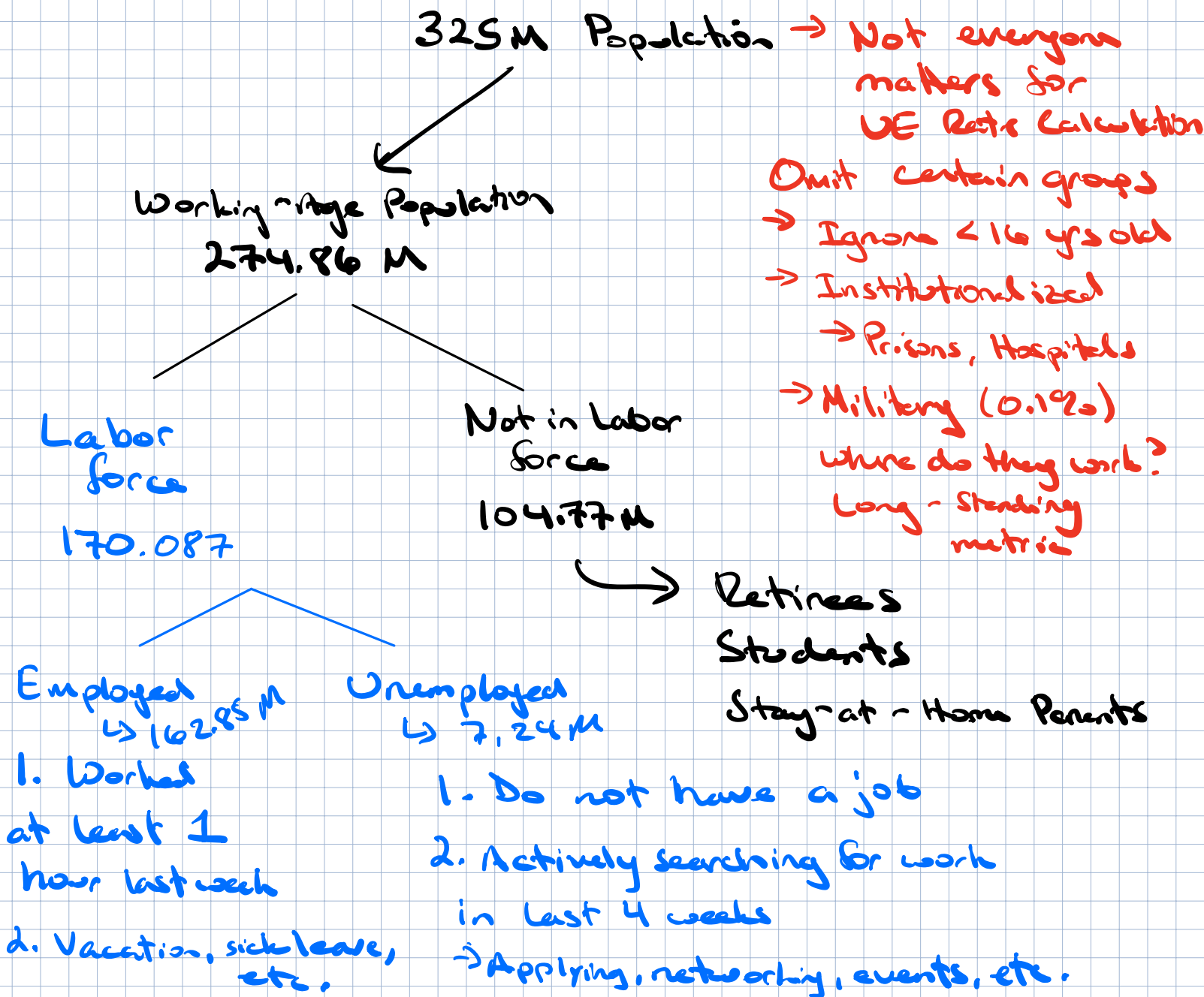
① Output decreases $\rightarrow$ unemployment rise

Cyclical UE

Goal = Full Employment = Cyclical UE = 0%

Full-Employment Level of Output = $\bar{Y}$ (Y -bar)
When cyclical UE = 0%

Calculation of UE Rate $\Rightarrow$ Bureau of Labor Stats



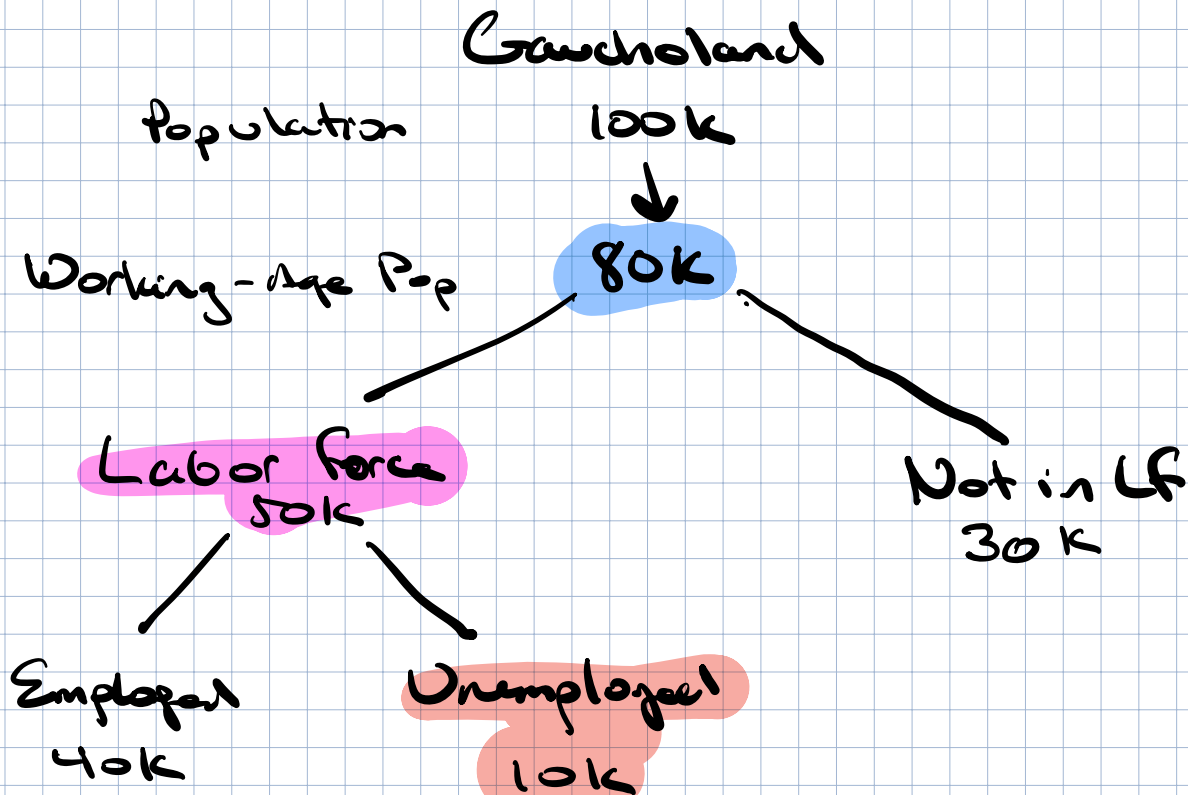
Unemployment Rate = What fraction of the labor market is unemployed?

$$\text{BLS Unemployed Rate} = \frac{\text{Number of UE}}{\text{Labor force}} \times 100$$

$$\text{March 2020 UE} = \frac{7.24 \text{ M}}{170.087 \text{ M}} \times 100 = 4.3\%$$

$$\text{Labor Force Participation Rate} = \frac{\text{Labor force}}{\text{Working-Age Pop}} \times 100$$

$$\text{LFP Rate} = \frac{170.087}{274.86} \times 100 = 61.9\%$$



$$\text{UE Rate} = \frac{10k}{50k} \times 100 = 20\%$$

$$\text{LFP Rate} = \frac{50k}{80k} \times 100 = 62.5\%$$

Shortcoming of UE Rate (BLS)

1. Individuals who work 1 hour / week
⇒ Fully employed

Do you want to work more hours?

Are you involuntarily part-time (IPT)?

Elliot ⇒ No job → wants 40 hours

Sutton ⇒ Has a job → work 20 hours, wants 40 hrs

Edwin ⇒ Has a job → work 20 hrs, want 40 hrs

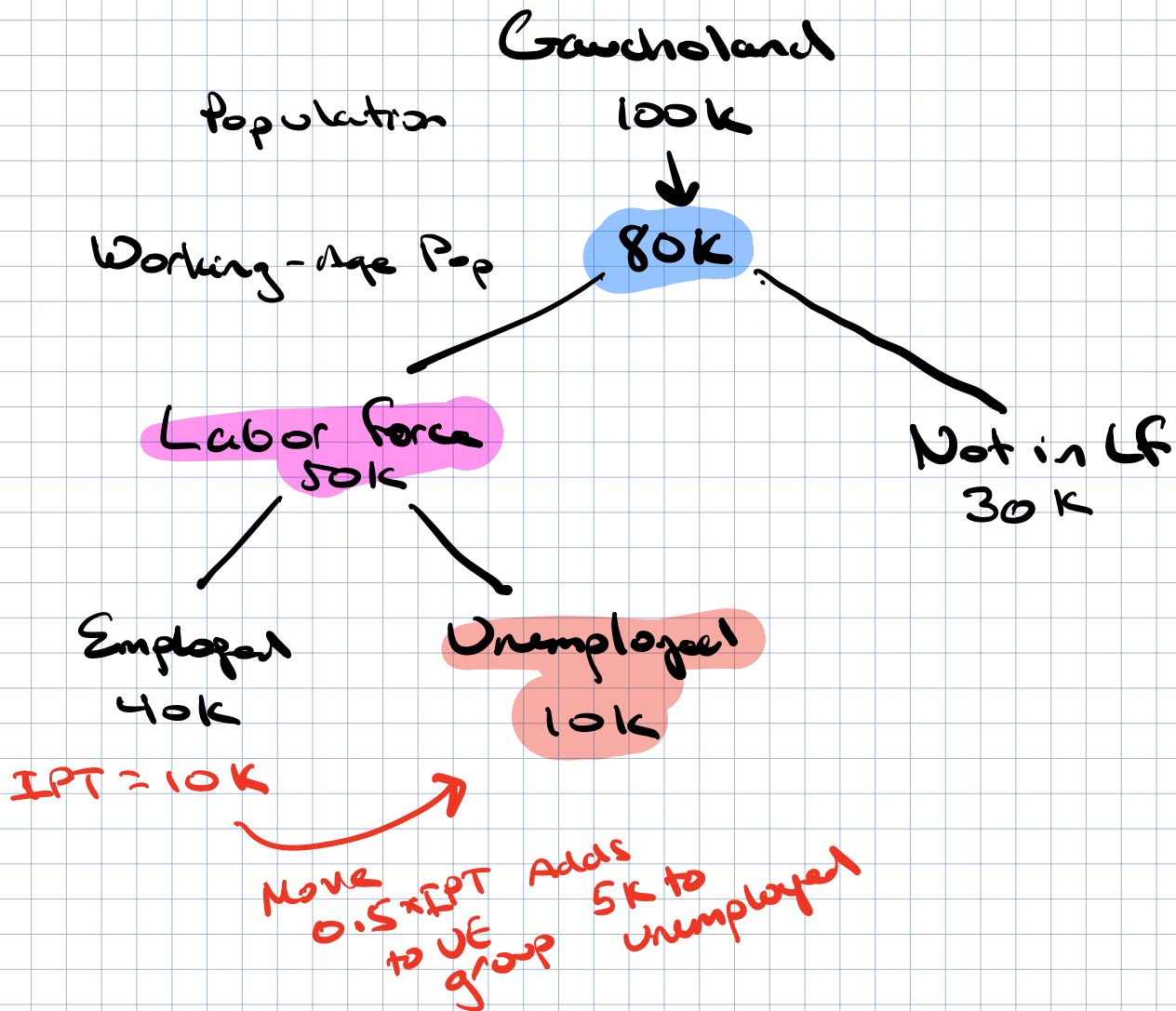
Sutton & Edwin together want 40 hrs more
⇒ Equivalent to what Elliot wants

IPT are partially unemployed → $\frac{1}{2}$ UE

Adjust original UE rate to consider IPT

$$\begin{aligned}\text{UE with IPT} &= \frac{\# \text{ of UE} + 0.5 \times \text{IPT}}{\# \text{ in Labor Force}} \times 100 \\ &= \frac{7.24 \text{ M} + 0.5 \times (4.5 \text{ M})}{170.087 \text{ M}} \times 100 \\ &= \frac{9.49 \text{ M}}{170.087 \text{ M}} \times 100 = 5.6\%\end{aligned}$$

Gaucholand $\neq$ 10K IPT workers $\rightarrow$ 1/2 unemployed



2. Actively searching definition is strict

Applied for a job in past 4 weeks?

Bulk applying 5 weeks ago & waiting?

Have you applied in past 12 months?

Discouraged Workers: not in labor force initially

$\rightarrow$ Haven't searched in past 4 weeks

$\rightarrow$ Searched in past year

$\rightarrow$ Take a job if offered

$$\begin{aligned}
 \text{UE Rate Adjusted for Discouraged Workers} &= \frac{\# \text{ of UE} + \# \text{ of Discouraged}}{\# \text{ in Labor force} + \# \text{ of Discouraged}} \times 100 \\
 &= \frac{7.24 \text{ M} + 5.73 \text{ M}}{170.087 \text{ M} + 5.73 \text{ M}} \times 100 \\
 &= \frac{12.97 \text{ M}}{175.82 \text{ M}} \times 100 \approx 7.4\%
 \end{aligned}$$

Gaucholand has 10k discouraged, adjusted UE Rate

